

Monitoring

Version: 1.0

Purpose

This document defines the monitoring strategy of Trading Platform Pro.

Monitoring provides continuous visibility into the health, performance and operational state of the application.

The objective is to detect problems early, simplify troubleshooting and support stable long-term operation.

Objectives

The monitoring system shall provide:

- runtime visibility
- health monitoring
- performance metrics
- operational diagnostics
- early problem detection
- historical analysis

Monitoring Principles

Monitoring should be:

- proactive
- lightweight
- deterministic
- non-intrusive
- continuously available

Monitoring must never change application behaviour.

Monitoring Scope

Typical monitored components include:

- Runtime
- Scheduler
- Event Processing
- Messaging
- Dependency Injection
- Configuration
- Plugins
- File Services
- Broker Integration
- Background Workers

Health Checks

Health checks provide the current operational state.

Typical checks:

- application startup
- configuration validity
- dependency container
- runtime services
- scheduler

- plugins
 - external services
- Health checks should execute quickly.
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Metrics

Typical runtime metrics include:

- application uptime
- startup duration
- memory usage
- CPU utilization
- event throughput
- queue length
- task execution time
- scheduler latency

Metrics should support trend analysis.

Alerting

Monitoring should detect:

- service failures
- repeated exceptions
- resource exhaustion
- unavailable dependencies
- abnormal execution times

Alerts should contain enough context for diagnosis.

Logging Integration

Monitoring complements logging.

Logging answers:

> What happened?

Monitoring answers:

> How healthy is the system?

Both systems should work together without duplication.

Performance Monitoring

Performance monitoring should identify:

- slow operations
 - bottlenecks
 - blocking tasks
 - excessive resource consumption
- Measurements should have minimal runtime overhead.
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Operational Dashboard

The monitoring infrastructure should support dashboards showing:

- runtime status
 - service health
 - active components
 - recent errors
 - key performance indicators
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Testing

Monitoring components should be independently testable.

Health checks and metrics should be validated through automated tests where practical.

Evolution

Future monitoring capabilities may include:

- distributed tracing
- centralized metrics
- notification channels
- historical trend analysis
- predictive diagnostics

Monitoring Review Checklist

Before release verify:

- health checks implemented
- critical metrics available
- alert conditions defined
- dashboards updated
- monitoring documented

Operational Readiness

Monitoring shall support production operation by providing:

- fast problem identification
- root cause analysis
- trend analysis
- service health visibility
- operational transparency

Monitoring should evolve together with the platform.

Related Documents

- Runtime.md
- Logging.md
- Configuration.md
- Infrastructure.md
- AGENTS.md